



stat()

Get file information.

```
#include <sys/stat.h>
#include <unistd.h>

struct stat fileStat;

int result = stat("/path/to/file.html", &fileStat);
```

What `stat()` provides:

- Field "st_mode" of type "mode_t" provides file type and permissions
- Field "st_size" of type "off_t" provides file size in bytes
- Field "st_mtime" of type "time_t" provides last modification time

Checking file type with "st_mode":

```
if (S_ISREG(fileStat.st_mode)) {
    // Regular file
}
else if (S_ISDIR(fileStat.st_mode)) {
    // Directory
}
else if (S_ISLNK(fileStat.st_mode)) {
```

```
    // Symbolic link (handle carefully)
}
```

Checking permissions:

```
// Check if readable by owner/group/other
if (fileStat.st_mode & S_IRUSR) {
    // Readable by owner
}
if (fileStat.st_mode & S_IRGRP) {
    // Readable by group
}
if (fileStat.st_mode & S_IROTH) {
    // Readable by others
}
```

<https://www.man7.org/linux/man-pages/man2/stat.2.html>